

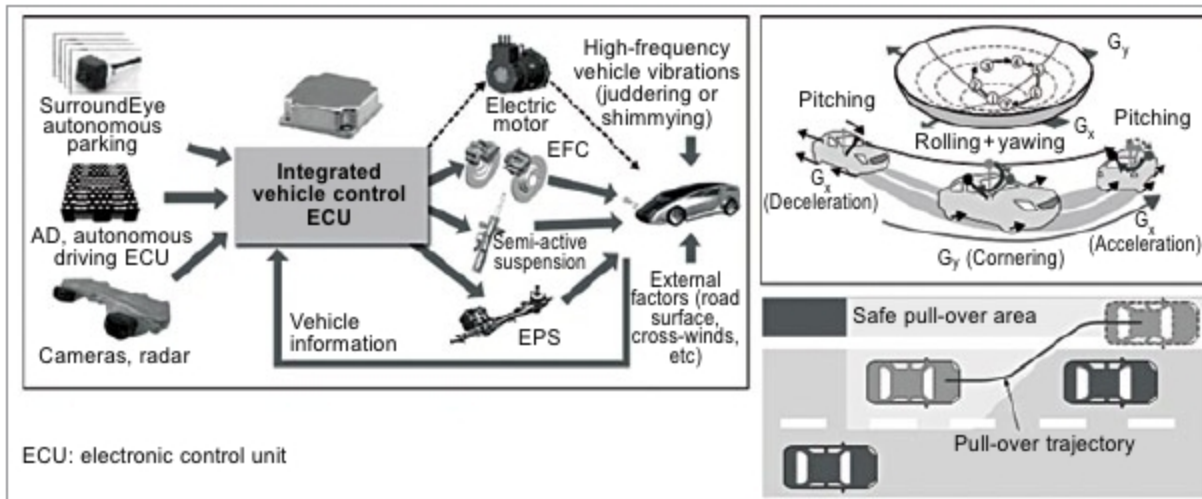


Fig. 2: Electronic control unit (Credit: www.hitachi.com)

which recreates continuous collision-free spaces, bounded by lanes and other obstacles between which the vehicle is expected to drive; uses data from digital maps built by simultaneous location and mapping models

Denver's (in Colorado, USA) Regional Transportation District has launched the first on-road deployment of an autonomous vehicle (shuttle). Although, it will not have a driver, an ambassador will always be present on the shuttle to ensure safety of the passengers and operations. Data collected will be used to improve future deployments and bring autonomous services into wider usage.

Design considerations for lidar

There are thermal challenges involved in designing lidar systems. The distance a lidar sensor can measure is affected by its operating temperature. For lidar systems that incorporate custom systems-on-chip (SoCs) ICs, operating temperature of internal components and external surfaces are influenced by their internal heat generation. Surrounding temperature and air flow, and cooling mechanisms are the key parameters to design lidar systems. Lidar systems that are mounted on vehicle roofs or other areas are directly exposed to the Sun. Hence, considerations for heating by solar radiation must be included.

To ensure that designs meet safety requirements, thermal simulation and analysis is done, which helps detect thermal issues before and in parallel with electrical and mechanical designing. This results in shortened overall design process and more robust product design. Thermal analysis done early in the design process helps select

designs that will work better among all other designs. With the development of electrical and mechanical designs, thermal aspects can also be refined, clearly indicating which designs should be prototyped and tested.

Good thermal management of PCBs and critical ICs on the board ensures reliability and good performance. Internal lidar components generate heat, which is convected, conducted and radiated to enclosure internal surfaces. Heat-transfer effectiveness for critical IC components can be improved by adding new conduction paths or through increasing heat transfer surface area. Using a heatsink on the device can lower component temperature.

A direct conduction path to the outer enclosure is recommended when a significant reduction in temperature is needed. Adding thermal conduction in solids such as aluminium is much more efficient than convection and conduction through air. Temperature reduction is influenced by IC heat dissipation and thermal resistance between the ICs and enclosure.

Digital design to speed up development

Thermal design optimisation for a lidar system analyses a wide range of vehicle integration and use case scenarios. Solar radiation can heat up vehicle-mounted lidar when a vehicle is being driven on a hot, sunny day. These simulation-driven analyses, from early prototyping stages, empower designers to account for various performance and life-limiting factors in their design and conduct optimisation analysis.

Testing autonomous vehicle func-

tionality in extremely hot and cold weather conditions is a critical part of vehicle verification and validation. To rely on the virtual design framework, accuracy of simulations is important for design engineers to make design decisions. Electronic cooling simulation can be calibrated with non-destructive thermal characterisation to ensure accuracy. Thermal simulation can be extremely useful to ensure reliability of lidar and other sensors, and autonomous functionality-enabling electronics for both sensor companies and vehicle manufacturers.

Integrated vehicle control structure

Lassaad Ben Cheikh, product marketing manager - automotive division, Infineon, says, "To meet safety and reliability requirements for autonomous vehicles, electronic systems must transition from fail-safe/fail-silent to fail-operating modes. To ensure robustness, availability and functional safety of critical systems, an effective strategy requires a two-channel architecture for electronic control units handling safety-critical functions." Safe automation levels in passenger vehicles defined by Society of Automotive Engineers (SAE) are shown in Fig. 3.

Due to integration of different components and automated features, it is important to ensure their interoperability within the vehicle. An autonomous car relies on devices and components that exchange, share and interpret data. Interoperability ensures that these devices and components can form an integrated ecosystem within the vehicle, seamlessly communicating with one another for a safe ride.

Interoperability of components ensures these will work together in a secure manner, without sacrificing performance. To check for issues as well as how other devices or components in a system impact each other, testers can place products in a simulated environment. Software testing is also an effective way to check interoperability as it uses manual and simulated test processes to verify that the product meets all functional, performance, security and quality requirements.